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6.1.1 Whose problem is it anyway?

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Homework 6.1.1.2

0.0/1.0 point (graded)

You use some linear system solver and it gives the wrong answer. In other words, you solve $Ax = b$ on a computer, computing $\hat{x}$, and somehow you determine that

$$\|x - \hat{x}\|$$

is large. Which of the following is a possible cause (identify all):

Mark all.

- ☐ There is a bug in the code. In other words, the algorithm that is used is sound (gives the right answer in exact arithmetic) but its implementation has an error in it.
- ☐ The linear system is ill-conditioned. A small relative error in the right-hand side can amplify into a large relative error in the solution.
- ☐ The algorithm you used accumulates a significant roundoff error.
- ☐ All is well: $\|\hat{x} - x\|$ is large but the relative error $\|\hat{x} - x\|/\|x\|$ is small.

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